

Jitin Gupta

Applied AI Architect

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SUMMARY

Applied AI Architect who designs AI systems that help investors see structural market shifts before prices reflect them. Leads architecture and product direction for an AI-assisted research and forecasting platform built for a global hedge fund with ~\$1B AUM, in close collaboration with a globally recognized expert on technology-driven economic disruption.

Combines agentic AI system design with quantitative modelling: causal models, cost and adoption curves, stock-flow dynamics and scenario analysis across energy, transport, batteries, commodities, AI infrastructure and labor. Sets direction for a 15+ person cross-functional team.

CORE CAPABILITIES

Applied AI & Agentic Systems: LLM applications, Agentic architectures, Workflow orchestration, Retrieval (RAG), Tool use, AI evaluation, Provenance, Prompt engineering, Anthropic Claude, OpenAI APIs

Quantitative Research & Forecasting: Causal modelling, Demand forecasting, Scenario & sensitivity analysis, Adoption S-curves, Cost & capability curves, Stock-flow modelling, Time-series analysis, Backtesting, Investment research

Product & Technical Leadership: AI product strategy, Research-to-engineering translation, Technical specifications, Systems design, Engineering review, Cross-functional leadership

Engineering & Tools: Python, SQL, Pandas, NumPy, scikit-learn, PyTorch, Git, Docker, REST APIs, AWS, Databricks

EXPERIENCE

CLOUDSUFI

Noida

Applied AI Architect

Jul 2025 – Present

Lead architecture and product direction for an AI-assisted research and forecasting platform that identifies structural market inflection points (technology-driven phase changes) before they are reflected in market prices. Built for a global hedge fund with ~\$1B AUM, in close collaboration with a globally recognized expert on technology-driven economic disruption.

- Architect agentic AI research workflows, defining how agents plan, retrieve context, call quantitative models and tools, verify outputs and synthesize answers.
- Define interfaces and responsibility boundaries between AI agents, forecasting models, data systems and evaluation layers, with provenance, reproducibility and explainability as design requirements.
- Lead a portfolio of 12+ sector forecasting models across energy, transport, batteries, commodities, AI infrastructure and labor, structured around cost and capability curves, adoption S-curves, stock-flow dynamics and scenario analysis.
- Translate investment hypotheses into causal models, assumptions, datasets and forecast outputs that the research team can test and challenge.
- Set product and technical direction across a 15+ person cross-functional team of AI/software engineers, research analysts and project management; turn research needs into specifications and engineering work packages.

Senior Software Engineer – Applied AI

Oct 2024 – Jun 2025

Hands-on engineering on the early versions of the AI research and forecasting platform for the hedge-fund engagement.

- Built LLM applications and prompt-chaining workflows that combined language models with quantitative forecasting models and research data.
- Developed and refined sector demand-forecasting models, turning domain hypotheses into data pipelines, model logic and scenario outputs.
- Worked directly with domain experts to turn research questions into model specifications and testable assumptions.
- Promoted to Applied AI Architect in July 2025.

Scaler

Remote

Instructor – Data Science

Aug 2023 – Oct 2024

- Taught working professionals Python, NumPy, Pandas, probability, statistics, mathematics for ML, supervised and unsupervised learning, neural networks and MLOps.
- Mentored learners through hands-on problem solving, explaining quantitative and ML concepts to audiences with mixed technical backgrounds.

Share India Securities

Gurgaon

Data Scientist – Algorithmic Trading

Apr 2023 – Jan 2024

- Researched quantitative trading strategies, including mean-reversion and machine-learning approaches, using historical OHLC market data and statistical methods.
- Built backtesting workflows that accounted for transaction costs, slippage, position sizing and entry/exit rules to test strategy robustness.
- Developed real-time trading workflows using market-data APIs and automated execution logic.
- Worked independently across the full cycle: research, data preparation, modeling, validation and implementation.

Vidyamandir Classes

New Delhi

Data Scientist

Apr 2020 – Mar 2023

- Built machine-learning models to predict student dropout risk and JEE Main scores, combining academic and behavioral data to guide retention and intervention decisions.
- Delivered an end-to-end prediction pipeline: dropout rates fell 20% and score predictions reached ~2% MAPE (mean absolute percentage error).
- Designed KPIs and dashboards so academic and business teams could track engagement, outcomes and operational performance.
- Presented model results to non-technical leadership and teaching teams.

Academic Operations Manager

Apr 2020 – Jan 2022

- Managed academic operations for 50+ educators serving 3,000+ students a year, and built training modules for 200+ employees across subjects and teaching methods.

Earlier roles

Independent Consultant – Web Analytics, Curate Partners	Jun 2024 – Oct 2024
Head of Physics Department, Vidyamandir Classes	Apr 2017 – Apr 2020
Senior Faculty, Vidyamandir Classes	May 2016 – Apr 2017
Co-founder & Business Development Manager, FoxBox	Jan 2015 – Jan 2016
Digital Marketing Manager, Loqation	Jan 2014 – Jan 2015

EDUCATION & CERTIFICATIONS

- Indian Institute of Technology Delhi, B.Tech., Civil Engineering (2007 – 2011)
- Claude Certified Architect – Foundations, Anthropic (Sep 2026, valid to Sep 2027)
- Databricks Certified Machine Learning Professional, Databricks (2024–2026, expired)
- Advanced Data Science and Machine Learning, Scaler Academy